

Too Big to Aggregate: Corporate Control and Policy-Invariant Aggregation

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National accounts record where economic activity takes place, while corporate control governs which activities adjust together. This paper asks which information a supplied macroeconomic state preserves for a declared policy response. Variation along the state directions omitted from that representation gives the minimum dimension of a local repair. Manifold-width geometry then measures the approximation cost of retaining fewer coordinates, with curvature bounds that extend the calculation to finite policy experiments. In euro-area manufacturing, ECB monetary responses vary across lagged corporate-control indices. The fitted sign pair appears in the policy-release window when release and conference shocks enter jointly and remains under an extended rate basis and conditioning on yield-curve, sovereign-spread, and pre-announcement-news components. For an admissible monetary decomposition and a quarter-unit index radius, omitting both measured corporate coordinates gives a fitted worst-case employment error of 0.1125 log percentage points across four horizons, with a simultaneous 95 percent interval of $[0.0312, 0.1939]$. Retaining one coordinate lowers the point error to 0.0165. At a benchmark of one million workers, these errors correspond to 1,125 and 165 local employment equivalents. The frontier varies with the outcome family, state metric, and policy scale. The empirical exercise concerns heterogeneity in monetary responses across measured corporate states.

1. Introduction

National accounts assign economic activity to the country where it takes place. Corporate groups make decisions across the activities they control. These two boundaries can describe the same current allocation while organizing its response to policy in different ways. Two economies with the same reported national state may therefore react differently to the same intervention. The state directions that generate this difference determine what must be added to the national representation and how costly a smaller addition would be.

The candidate missing state is the corporate decision perimeter. Common control links establishments through internal funding, shared adjustment costs, and joint allocation, while national accounts continue to record their activity by location. The paper asks how much information about monetary responses is left out when a national state omits measured control geography, and how many control coordinates are needed to recover a declared target.

This is an invariance question in the sense of Marschak (1953); Hurwicz (1962); Lucas (1976): which relationships remain usable under a stated class of interventions? Marschak's Maxim directs identification toward the objects needed for the policy question (Heckman 2010). Here the object supplied by existing institutions is a state representation q_R . The question is which intervention-relevant information that representation preserves. The policy, outcomes, horizons, and base state together define the target $T_{\mathcal{P}}$ and, with it, the information required for the exercise.

Consider two countries with the same employment, capital, production, and establishments, but with different corporate boundaries. A firm operating across countries can reallocate funds and share adjustment costs across its activities. A monetary contraction can then reach identical local balance sheets through different joint decisions. Common-control curvature, shared inputs, and strategic feedback enter the inverse equilibrium Jacobian that maps policy incidence into national outcomes.

The analysis begins with the hidden-response operator

$$\mathcal{L}_{\mathcal{P}} = DT_{\mathcal{P}}|_{\ker Dq_R}.$$

Linear factorization gives a first-order closure criterion, and the rank of this operator gives the minimum dimension of a local augmentation. Theorem 4 then uses classical manifold-width geometry (DeVore et al. 1989; Dahlke et al. 2006) to obtain the first-order compression cost $|\tau|\rho\sigma_{d+1}$ when d continuous coordinates are retained. The policy displacement τ , state radius ρ , target weights, and state metric give this cost its economic units. Curvature and projection-discrepancy bounds carry the calculation to a finite policy experiment. Together, these objects describe what the supplied representation leaves unresolved for a particular policy target and how much of that information a smaller repair can recover.

The closest comparison comes from the aggregation literature. Krusell and Smith (1998) establish approximate aggregation using the mean of the wealth distribution, while Den Haan and Rendahl (2010) connect individual policy rules to the aggregate state variables they require. An et al. (2009); Chang et al. (2013) study representative-agent descriptions and policy-dependent aggregate parameters. Koby and Wolf (2020) express general-equilibrium smoothing through the rank deficiency and null space of a dynamic demand–supply elasticity map, which determines which distribution-induced demand shifts survive price adjustment. The analysis here starts from an existing statistical representation and restricts attention to the directions in $\ker Dq_R$. The declared response target then determines the dimension and loss frontier of the repair.

A related literature asks which information is sufficient for policy evaluation. Forecasts and impulse responses enter the policy assessment of Barnichon and Mesters (2023), and causal monetary responses support the near-term counterfactuals of Caravello et al. (2026). This paper asks which current-state information preserves those response functions. Sequence-space Jacobians provide a macroeconomic language for equilibrium derivatives (Auclert et al. 2021); the concern here is the state information on which those derivatives depend.

The empirical analysis pairs lagged Eurostat FATS control indices with intraday ECB surprises and conditions on eight national variables and their monetary and information interactions. The central admissible rotation gives four-quarter employment loadings $(-0.466, 0.631)$. The estimand is heterogeneity in monetary responses across measured corporate states. Assignment of corporate control requires a separate design, and the Appendix reports evidence from within-country and issuer data. Following Chen and Fang (2019), the rank analysis allows for the rank-zero stratum. The comparison across methods leads naturally to a repair frontier reported with sampling uncertainty.

For a quarter-unit index radius and a one-event-standard-deviation monetary innovation, the fitted employment RMS frontier is $(0.1125, 0.0165, 0)$ when zero, one, and two coordinates are retained. The simultaneous operator-norm interval for the zero-coordinate loss is $[0.0312, 0.1939]$. Retaining the first coordinate lowers the point loss by 0.0960, or 85.3 percent. Direct evaluation on the aligned score draws gives a 95 percent basic interval of $[0.0286, 0.1598]$ for this reduction and a remaining-error interval of $[0, 0.0979]$. When credit and investment enter with fixed outcome-standard-deviation weights, the share attributed to the second direction rises from 2.11 to 14.37 percent. The usefulness of an additional coordinate therefore depends on the target it is meant to preserve.

Studies of multinational measurement reassign financial positions or profits to trace their aggregate consequences (Coppola et al. 2021; Guvenen et al. 2022). My exercise holds the current allocation fixed and lets the decision perimeter change its response. Evidence on internal funding and international transmission provides the empirical basis

for this mechanism (Biermann and Huber 2024; Cetorelli and Goldberg 2012; Zhang et al. 2025). The FATS indices describe the geography of corporate control, while the numerical economy shows how joint decisions generate the hidden-response directions.

2. Corporate Control and the Aggregation Problem

2.1. States, activities, and national statistics

There are C countries and N corporate groups. Group i controls a vector $a_i = (a_{i1}, \dots, a_{iC})'$ of local pricing, production, capacity, investment, financing, or sourcing decisions. The full state $S_t \in \mathcal{S}$ records technologies, local stocks, ownership links, corporate geography, financial positions, shared inputs, and strategic connections. On the neighborhoods considered below, the state space $\mathcal{S}$ and the national-state space $\mathcal{X}$ are finite-dimensional C^2 manifolds. Existing statistical institutions provide the map

$$(1) \quad X_t = q_R(S_t) \in \mathcal{X}.$$

Thus X_t contains the information available in a residence- or location-based macroeconomic representation.

Contemporaneous national outcomes $Y_t \in \mathbb{R}^{d_Y}$ take the local form

$$(2) \quad Y_t = Ma_t + y(X_t),$$

where the stacked action has dimension m and $M \in \mathbb{R}^{d_Y \times m}$ is fixed by the accounting convention. Dynamic targets propagate through

$$(3) \quad X_{t+1} = \mathcal{G}(X_t, Y_t, \xi_{t+1}).$$

For a policy p and horizon H , $T_p(S_t)$ stacks the conditional paths of the declared outcomes. The policy family and target vector are fixed before estimation.

2.2. The corporate decision problem

Firm i maximizes

$$(4) \quad \Pi_i(a_i, a_{-i}; S, p) = R_i(a_i, a_{-i}; S, p) - \frac{1}{2} a_i' D_i a_i - \frac{1}{2} \sum_{k=1}^{r_i} \lambda_{ik} (g_{ik}' a_i)^2.$$

The matrix D_i is symmetric positive semidefinite and $\lambda_{ik} \geq 0$. Each loading g_{ik} joins activities that use a common intangible, managerial capacity, internal liquidity, product architecture, data infrastructure, or another firm-wide resource. Reassigning a local

activity changes which cross-country terms enter the owner's decision problem, even when current local capital, employment, output, and sales remain fixed.

Stack equilibrium conditions as

$$(5) \quad F(a; S, p) = 0, \quad J(S, p) = D_a F(a; S, p).$$

The equilibrium Jacobian can be written as

$$(6) \quad J = H^L + H^C + H^G - \Gamma.$$

The block H^L collects local curvature, H^C common-control effects, H^G shared-input and reallocation effects, and Γ strategic cross-firm feedback. These blocks are determined by ownership, operating, and network objects measured before the policy exercise.

2.3. Three organizational channels

Common control. An owner internalizes the effect of one affiliate's decision on the returns to other controlled activities. A control change alters the off-diagonal decision curvature even when every activity remains in the same physical location.

Shared inputs and reallocation. A firm can move liquidity, managerial attention, intangible capital, R&D knowledge, capacity, or product design across its controlled locations. A policy change can therefore move local activities through an internal shadow price that the national-accounts state does not record.

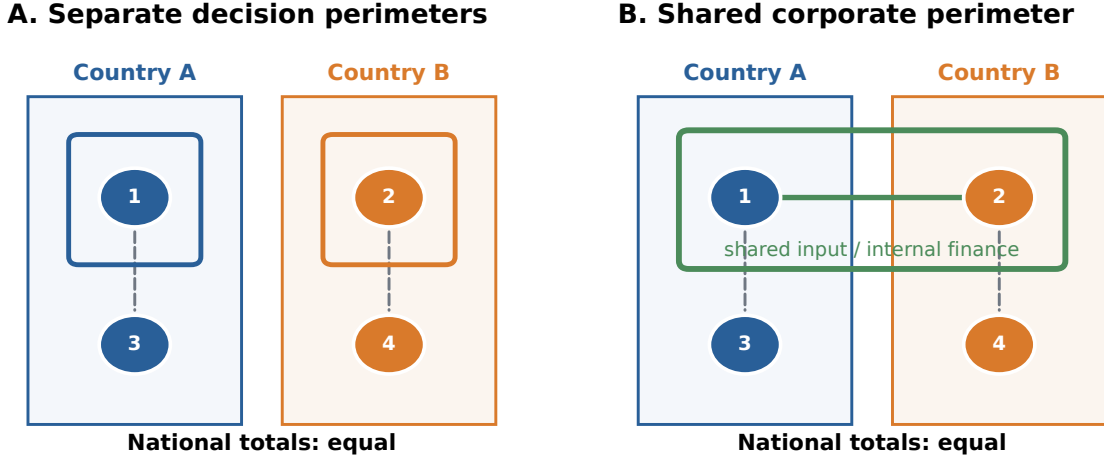
Strategic propagation. Competitors respond to the combined product and geographic footprint of the corporate group. A change in control changes the best-response system faced by rivals and can propagate an initially local policy exposure through markets or networks.

These three channels describe how corporate control can change the equilibrium response operator. For the declared policy family, the national aggregate remains sufficient when the derivatives of all three channel blocks vanish along the hidden control directions.

2.4. Currency-area exposure and geographic buffers

Let B^F measure the foreign-controlled share of world-classified local activity and let B^E measure the extra-EU-controlled share of world-classified local activity. Both enter the empirical implementation as within-vintage standardized indices averaged across four activity margins. A contractionary monetary innovation $\varepsilon_t > 0$ raises the affiliate financing wedge by $(1 + \alpha_H B^F - \gamma_H B^E) \varepsilon_t$, where $\alpha_H > 0$ captures transmission through a group-wide

FIGURE 1. Same National State, Different Corporate Control



The same residence-based allocation can map to different decision Jacobians.

Each panel holds the residence-based allocation fixed. The green perimeter in panel B joins activities 1 and 2 inside one decision problem and introduces a shared-input or internal-finance link.

euro balance sheet and $\nu_H > 0$ captures access to cash flow and financing outside the currency area. Let cumulative labor adjustment solve

$$(7) \quad \min_{\Delta n_{t:t+H}} \left\{ \frac{(\Delta n_{t:t+H})^2}{2\kappa_H} + [\ell_{0H} + (1 + \alpha_H B^F - \nu_H B^E) \varepsilon_t] \Delta n_{t:t+H} \right\}, \quad \kappa_H > 0.$$

PROPOSITION 1 (Two-coordinate employment transmission). *The solution to equation (7) satisfies*

$$(8) \quad \frac{\partial^2 \Delta n_{t:t+H}}{\partial \varepsilon_t \partial B^F} = -\kappa_H \alpha_H < 0, \quad \frac{\partial^2 \Delta n_{t:t+H}}{\partial \varepsilon_t \partial B^E} = \kappa_H \nu_H > 0.$$

The pair (B^F, B^E) spans the financing block whenever the two empirical loadings are linearly independent across declared outcomes or horizons.

The first derivative captures amplification through the part of the corporate perimeter exposed to the common policy innovation. The second captures insulation through resources whose cash flows and financing conditions extend beyond the currency area. Equation (8) gives sign restrictions for the primary employment response. The general Jacobian system allows common-input and strategic channels to operate alongside this financing mechanism.

3. A Theory of Policy-Valid Aggregation

3.1. Regularity and hidden control perturbations

Fix a baseline (a_0, s_0, p_0) satisfying $F(a_0; s_0, p_0) = 0$. The analysis uses three benchmark assumptions.

ASSUMPTION 1 (Smooth regular equilibrium). $F : \mathbb{R}^m \times \mathcal{S} \times \mathbb{R}^q \rightarrow \mathbb{R}^m$ is C^2 near the baseline and $J_0 = D_a F(a_0; s_0, p_0)$ is nonsingular. The local equilibrium supplied by the implicit-function theorem is denoted $a(s, p)$, and $J(s, p) := D_a F(a(s, p); s, p)$.

ASSUMPTION 2 (Institutionally fixed aggregation). q_R is C^1 and has constant rank near s_0 . The accounting map M in equation (2) is fixed on that neighborhood.

ASSUMPTION 3 (Pure control perturbations). There is a linear space $\mathcal{V}_C(s_0) \subseteq \ker Dq_R(s_0)$ such that every $h \in \mathcal{V}_C(s_0)$ satisfies $D_h M = 0$, $D_h F_p = 0$, and $D_h H^L = 0$. The perturbation may change H^C , H^G , and Γ .

The state-direction derivatives of J , M , and F_p below are total derivatives evaluated along $a(s, p)$. Assumption 3 isolates propagation through the organization of the firm. If accounting weights or direct policy incidence also vary with control, the response derivative contains the corresponding additional terms. A direct target component $G(S, p)$ contributes $D_h G_p$.

The implicit-function theorem gives

$$(9) \quad D_p a = -J^{-1} F_p, \quad R_p = D_p Y = -MJ^{-1} F_p.$$

THEOREM 1 (Control transmission). Under assumptions 1–3, every hidden control perturbation $h \in \mathcal{V}_C(s_0)$ changes the policy response by

$$(10) \quad \begin{aligned} D_h R_p &= MJ^{-1} (D_h J) J^{-1} F_p \\ &= MJ^{-1} D_h H^C J^{-1} F_p + MJ^{-1} D_h H^G J^{-1} F_p - MJ^{-1} D_h \Gamma J^{-1} F_p. \end{aligned}$$

For a finite menu of policy directions $v_1, \dots, v_K \in \mathbb{R}^q$, define the baseline incidence matrix

$$P = [F_p v_1, \dots, F_p v_K].$$

The entire family is locally insensitive to h if and only if $MJ^{-1} (D_h J) J^{-1} P = 0$.

The three factors in equation (10) trace the path from a change in organization to a change in the national response. The rightmost factor selects the decision margins moved by policy. The middle factor changes the equilibrium operator, and the leftmost factor

records what reaches the aggregate target. A hidden control change matters for the target when it survives each of these maps.

PROPOSITION 2 (Sufficient conditions for aggregation). *If $D_h H^C = D_h H^G = D_h \Gamma = 0$ for every $h \in \mathcal{V}_C(s_0)$, then $D_h R_p = 0$ for every policy in the declared family.*

3.2. Primitive response dimension

To relate this derivative to a finite set of measurable coordinates, suppose that organizational effects are generated by a fixed number of primitives.

ASSUMPTION 4 (Finite control-factor structure). *There are fixed matrices $B_1, \dots, B_R \in \mathbb{R}^{m \times m}$ and predetermined linear functionals $\ell_r : \mathcal{V}_C(s_0) \rightarrow \mathbb{R}$ such that*

$$(11) \quad D_h J = \sum_{r=1}^R \ell_r(h) B_r.$$

Let $L_C(h) = (\ell_1(h), \dots, \ell_R(h))'$ and define $A_{\mathcal{P}}$ by stacking $\text{vec}(MJ^{-1}B_r J^{-1}P)$ as columns. Thus $A_{\mathcal{P}} : \mathbb{R}^R \rightarrow \mathbb{R}^{d_Y K}$, $L_C : \mathcal{V}_C(s_0) \rightarrow \mathbb{R}^R$, and the vectorized hidden response $\mathcal{H}_{\mathcal{P}} : \mathcal{V}_C(s_0) \rightarrow \mathbb{R}^{d_Y K}$ have fixed domains and codomains.

THEOREM 2 (Primitive hidden-response dimension). *The hidden response factors as*

$$(12) \quad \mathcal{H}_{\mathcal{P}} = A_{\mathcal{P}} L_C,$$

and

$$(13) \quad \text{rank}(\mathcal{H}_{\mathcal{P}}) \leq \min\{\text{rank}(A_{\mathcal{P}}), \text{rank}(L_C)\} \leq R.$$

If $A_{\mathcal{P}}$ has full column rank R and L_C is onto $\mathbb{R}^R$, then $\text{rank}(\mathcal{H}_{\mathcal{P}}) = R$.

Assumption 4 is a maintained restriction on the economic model. The resulting upper bound applies to the control subspace specified there. Extending that bound to the full national kernel requires the spanning and exclusion conditions in corollary 1. An empirical matrix with two columns has rank at most two by construction, so its column count describes the dimension of the measured projection. An economic claim about the full kernel requires the additional conditions.

3.3. Closure and minimum repair

Let $T_{\mathcal{P}} : \mathcal{S} \rightarrow \mathbb{R}^{d_T}$ collect the declared target vector and define the national vertical space

$$(14) \quad \mathcal{V}_R(s_0) = \ker Dq_R(s_0).$$

DEFINITION 1 (First-order counterfactual closure). *The national representation is first-order closed for $T_{\mathcal{P}}$ at s_0 if there exists a linear map $A : T_{x_0}\mathcal{X} \rightarrow \mathbb{R}^{d_T}$ such that $DT_{\mathcal{P}}(s_0) = ADq_R(s_0)$.*

PROPOSITION 3 (Local national aggregation). *Suppose q_R and $T_{\mathcal{P}}$ are C^1 and q_R has constant rank near s_0 . First-order closure holds at s_0 if and only if*

$$(15) \quad \ker Dq_R(s_0) \subseteq \ker DT_{\mathcal{P}}(s_0).$$

If the inclusion holds throughout a sufficiently small constant-rank neighborhood U with connected local vertical sections, there is a C^1 map $\bar{T}_{\mathcal{P}} : q_R(U) \rightarrow \mathbb{R}^{d_T}$, where $q_R(U)$ carries its constant-rank image-manifold structure, such that $T_{\mathcal{P}}|_U = \bar{T}_{\mathcal{P}} \circ q_R|_U$.

The derivative condition is pointwise. The constant-rank neighborhood in the second part specifies where it can be integrated into a local factorization. Section 3.4 treats a different question: how much error remains under finite state and policy displacements when the representation is only approximately repaired.

Define the hidden-response operator and its rank by

$$(16) \quad \mathcal{L}_{\mathcal{P}}(s_0) = DT_{\mathcal{P}}(s_0)|_{\mathcal{V}_R(s_0)}, \quad r_{\mathcal{P}}(s_0) = \text{rank } \mathcal{L}_{\mathcal{P}}(s_0).$$

An augmentation $b : \mathcal{S} \rightarrow \mathbb{R}^d$ is a first-order repair if (q_R, b) satisfies the kernel criterion.

THEOREM 3 (Minimum control-state dimension). *Every d -dimensional C^1 first-order repair satisfies*

$$(17) \quad d \geq r_{\mathcal{P}}(s_0).$$

Conversely, there is a C^1 augmentation b^ defined on a neighborhood of s_0 and valued in $\mathbb{R}^{r_{\mathcal{P}}(s_0)}$ whose differential at s_0 separates exactly the quotient $\mathcal{V}_R(s_0)/\ker \mathcal{L}_{\mathcal{P}}(s_0)$ and makes (q_R, b^*) first-order closed at s_0 .*

COROLLARY 1 (Primitive rank-dimensional repair). *Suppose $\mathcal{H}_{\mathcal{P}}$ is a coordinate block of $DT_{\mathcal{P}}(s_0)|_{\mathcal{V}_C(s_0)}$ and has rank R . Every repair then has dimension at least R . This bound is attained by a predetermined C^1 control map $b_0 : \mathcal{S} \rightarrow \mathbb{R}^R$ when there is a direct-sum decomposition $\mathcal{V}_R(s_0) = \mathcal{V}_C(s_0) \oplus \mathcal{V}_O(s_0)$ satisfying $Db_0|_{\mathcal{V}_C} = L_C$, $Db_0|_{\mathcal{V}_O} = 0$, $DT_{\mathcal{P}}|_{\mathcal{V}_O} = 0$, and $\ker L_C \subseteq \ker(DT_{\mathcal{P}}|_{\mathcal{V}_C})$.*

The proofs use kernel inclusion and rank-nullity, but the objects entering those arguments are economic. The map q_R is fixed by the statistical institution, $T_{\mathcal{P}}$ is fixed by the policy exercise, and the corporate primitives determine how hidden control directions change J^{-1} . A repair built directly from the response coordinates uses the target derivative

itself. Interpreting measured control coordinates as that repair requires the spanning conditions in corollary 1.¹

3.4. Approximate repair and finite policy changes

Suppose the policymaker can add d coordinates to the reported state. The remaining question is how much counterfactual accuracy those coordinates can preserve. Fix a constant-rank chart $s(z)$ of a national fiber, with $q_R(s(z)) = x_0$ and $z \in \bar{B}_\rho \subset \mathbb{R}^k$. This chart fixes the units in which hidden-state variation is measured. Let $Y(z, \tau)$ collect the declared outcomes and horizons under a scalar policy displacement τ along a fixed direction, and define the policy increment $G(z, \tau) = Y(z, \tau) - Y(z, 0)$. Write $R(z) = \partial_\tau Y(z, 0)$ and $L = D_z R(0)$. On this fiber, L is the hidden-response operator for the target $T_{\mathcal{P}} = R$. Equip the state and target spaces with specified Euclidean norms, and order the singular values of L as $\sigma_1(L) \geq \dots \geq \sigma_k(L) \geq 0$, padding with zeros where needed.

ASSUMPTION 5 (Uniform finite-experiment regularity). *The fiber chart and Y extend to an open neighborhood of $\bar{B}_\rho \times [-\bar{\tau}, \bar{\tau}]$. The response R is twice continuously differentiable, Y is twice continuously differentiable in τ , and throughout that set $\|D_z^2 R(z)\| \leq K_s$ and $\|\partial_\tau^2 Y(z, \tau)\| \leq K_p$. The first bound is the bilinear operator norm.*

For $0 \leq d < k$, let a continuous state encoder $b : \bar{B}_\rho \rightarrow \mathbb{R}^d$ supply the additional coordinates, and let g be any decoder. Define the best attainable error at policy displacement τ by

$$(18) \quad e_d(\rho, \tau) = \inf_{b \text{ continuous}, g} \sup_{\|z\| \leq \rho} \|G(z, \tau) - g(b(z), \tau)\|, \quad \epsilon(\rho, \tau) = \frac{|\tau|K_s\rho^2 + K_p\tau^2}{2}.$$

THEOREM 4 (Policy-response compression frontier). *Under assumption 5, for $\rho > 0$ and $|\tau| \leq \bar{\tau}$,*

$$(19) \quad \max\{0, |\tau|\rho\sigma_{d+1}(L) - \epsilon(\rho, \tau)\} \leq e_d(\rho, \tau) \leq |\tau|\rho\sigma_{d+1}(L) + \epsilon(\rho, \tau).$$

The upper bound is attained as a bound by the leading right-singular coordinates $b_d(z) = (v'_1 z, \dots, v'_d z)$ and decoder $g_d(b_d(z), \tau) = \tau R(0) + \tau \sum_{j=1}^d \sigma_j u_j v'_j z$. For the first-order experiment $G(z, \tau) = \tau[R(0) + Lz]$, the equality $e_d = |\tau|\rho\sigma_{d+1}(L)$ holds even over continuous nonlinear encoders and arbitrary decoders. For $d \geq k$, an identity encoder and the exact response decoder give $e_d = 0$.

¹Xon (2026) develops source-record sufficiency, common measurement coordinates, and certificate choice for revaluing the same economic object. The shared fiber-factorization geometry supplies a common tool. Here it is applied to policy-response aggregation, corporate equilibrium transmission, and finite counterfactual compression.

The first-order equality is the finite-dimensional Hilbert-space case of nonlinear and manifold widths (DeVore et al. 1989; Dahlke et al. 2006). The Bernstein-width lower bound uses antipodal coincidence (Borsuk 1933), while singular-value truncation attains the upper bound (Eckart and Young 1936). Standard manifold widths require both the encoder and decoder to be continuous. Here the lower bound uses continuity of the encoder, and the linear decoder that attains the bound is itself continuous, so the two decoder classes have the same value on this ellipsoid.

In the economic application, this width measures response information discarded by q_R . The radius ρ , policy displacement τ , state and target metrics, and curvature budget determine its units and the range over which the finite bound is informative. For $\tau \neq 0$, the lower bound is positive when $\sigma_{d+1} > K_s \rho / 2 + K_p |\tau| / (2\rho)$. Baseline allocations enter separately through $Y(z, 0)$.

The curvature constants can also be read through the corporate decision perimeter. On a smooth fiber where $R(z) = -MJ(z)^{-1}P$, suppose M and P are fixed and $\|J^{-1}\| \leq \kappa$, $\|D_z J\| \leq J_1$, and $\|D_z^2 J\| \leq J_2$. Then one may take

$$(20) \quad K_s = \|M\| \|P\| (2\kappa^3 J_1^2 + \kappa^2 J_2).$$

The inverse equilibrium Jacobian amplifies the hidden first-order response as well as the finite-state remainder. Policy-dependent strategic feedback also enters K_p , as shown in the numerical illustration. Section 5.4 reports the frontier in employment units. Corollary A1 carries the bound from a measured-index map to an economic fiber by accounting for metric distortion and response-projection discrepancy.

3.5. Policy menus and target scope

Adding policies or horizons stacks rows in the hidden-response system, so the rank can only increase. Each rank statement therefore refers to a particular policy, set of outcomes, horizons, sample window, and primitive upper bound. A state that is sufficient for one monetary experiment may require more coordinates for a family that also includes fiscal, regulatory, or cross-border policies. Firm-level response images that point in different directions can raise the dimension of the aggregate repair. If all firms share an r -dimensional response image, the aggregate rank remains bounded by r .

If q_R assigns the same reported state to s and s' while $\|T_{\mathcal{P}}(s) - T_{\mathcal{P}}(s')\| = \Delta > 0$, any deterministic predictor based on that report has worst-case error of at least $\Delta/2$. The quantitative collision exercise expresses this separation in economic units.

4. Measuring Corporate Control

The application studies euro-area manufacturing under ECB monetary innovations. Eurostat inward FATS assigns the residence of control to the ultimate controlling institutional unit (Eurostat 2026). Control is defined through authority over an affiliate’s general policy, including the appointment of directors. This measure describes control geography; internal liquidity sharing and operational integration require separate evidence. For each manufacturing margin m , let W_{mct} denote world-classified local activity, F_{mct} foreign-controlled activity, and E_{mct} extra-EU-controlled activity. The primitive shares are

$$(21) \quad s_{mct}^F = F_{mct}/W_{mct}, \quad s_{mct}^E = E_{mct}/W_{mct}.$$

Employment, tangible investment, turnover, and value added each provide one margin. The two indices average the corresponding within-vintage standardized shares and require at least three margins in the closed unit interval. A policy observation in year t is paired with the index from $t - 1$. The primary window, 2009Q2–2020Q4, uses the legacy production system throughout. Appendix B records the EBS extension, source flags, and transformation dictionary.

The state file contains 222 country-years across the two production vintages, and the two indices have a correlation of 0.780. Among the 220 observations with all four margins in both coordinates, the three split-half correlations range from 0.954 to 0.968. First-principal-component scores correlate above 0.9999 with the equal-weight composites. These calculations describe internal consistency. They leave room for source errors shared across the four margins. Appendix B reports margin-exclusion estimates under the alternative shock benchmark. The F/W and E/W indices measure activity-weighted control geography. The conditional share $g^{E|F} = E/F$, where $E/F = (E/W)/(F/W)$, is used only as a separate illustration moment.

The monetary data contain 315 EA-MPD event dates from January 1999 through October 2025 (Altavilla et al. 2019). EA-MPD reports a policy-release window, a press-conference window, and the union of the two. This union is the main treatment. A short-rate factor formed from one-, three-, six-, and twelve-month OIS changes explains 82.70 percent of complete-event variance. Contractionary monetary news raises rates and lowers equities, while information news raises both (Jarociński and Karadi 2020). Orthogonal sign-restricted components follow the construction in Arias et al. (2018). The reference component uses the midpoint angle -0.691244 in the admissible interval $(-1.382489, 0)$ determined by event covariance and the sign restrictions. The shocks have unit event variance, and quarterly shocks sum the event contributions. Section 5.5 considers the full admissible set. The Appendix labels the alternative standardized prior-mean shock used in the robustness specifications.

5. Do National Aggregates Close Monetary Policy Responses?

5.1. National benchmark state

The national benchmark contains eight quarter-end variables: lagged manufacturing-production growth, headline inflation, energy inflation, released GDP growth, NFC credit growth, the NFC lending rate, trade openness, and the manufacturing value-added share. Each enters in levels and in interactions with the monetary and information innovations. Country and calendar-quarter effects complete the benchmark. Write $q_N(S_{ct})$ for this observed representation. The regression asks whether the two corporate coordinates add information to the conditional response projection generated by q_N .

In the theory, q_R denotes the national state available to the statistical institution. The empirical projection based on q_N transfers to that broader object when changes in the indices admit a lift into $\ker Dq_R$ and the response variation associated with national coordinates omitted from q_N is bounded by the projection discrepancy in corollary A1. The candidate corporate state consists of the one-year-lagged foreign-control and extra-EU-control indices. The primary window ends in 2020Q4, which keeps every state coordinate within the legacy FATS production system.

5.2. Conditional response closure

For country c , quarter t , and horizon h , estimate

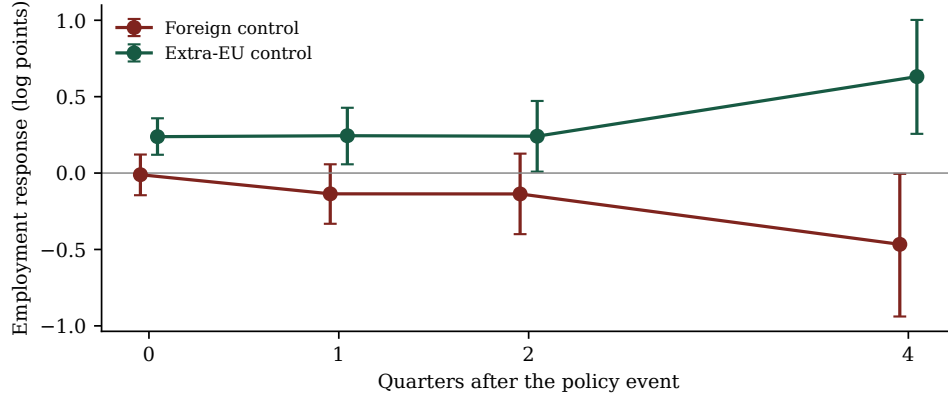
$$(22) \quad \begin{aligned} \Delta_h n_{ct} = & \alpha_{c,h} + \tau_{t,h} + \sum_{k=1}^8 \left(\theta_{kh} X_{kct} + \omega_{kh} X_{kct} \varepsilon_t^P + \zeta_{kh} X_{kct} \varepsilon_t^I \right) + g_h(B_{c,t-1}) \\ & + \delta_{Fh} B_{c,t-1}^F \varepsilon_t^P + \delta_{Eh} B_{c,t-1}^E \varepsilon_t^P \\ & + \rho_{Fh} B_{c,t-1}^F \varepsilon_t^I + \rho_{Eh} B_{c,t-1}^E \varepsilon_t^I + \eta_{c,t+h}. \end{aligned}$$

Here $\Delta_h n_{ct}$ is cumulative manufacturing-employment log growth from t through $t + h$, ε_t^P is the contractionary monetary component, and ε_t^I is the information component. Calendar-quarter effects absorb the common shock level. The state-dependent monetary response is $(\delta_{Fh}, \delta_{Eh})$, and proposition 1 predicts $\delta_{Fh} < 0$ and $\delta_{Eh} > 0$. The horizon-specific restriction is

$$(23) \quad H_0 : \quad \delta_{Fh} = \delta_{Eh} = 0.$$

Rejecting equation (23) means that the state directions summarized by the two corporate coordinates shift the declared policy response conditional on the national benchmark. The intraday policy window provides temporal assignment. Predetermined state coordinates, national-state shock interactions, and information-shock interactions form the condition-

FIGURE 2. Corporate Control and the Employment Response



Points use the central admissible monetary decomposition on the common 47-quarter support of the twelve-row system. Bars are pointwise 95 percent basic intervals from 9,999 aligned wild calendar-quarter draws. Every equation includes the eight national variables in levels and interacted with both shocks. Employment is cumulative $100\Delta \log N$ from quarter t through horizon h . These intervals condition on the measured state and event covariance.

ing structure. Under the event-timing exclusion, sign-decomposition, and common-shock assumptions, ε_t^p is the assigned treatment and $(\delta_{Fh}, \delta_{Eh})$ traces its conditional response surface across corporate states.

At four quarters, a one-unit increase in the foreign-control index lowers the response to a unit monetary innovation by 0.466 log percentage points. The extra-EU loading is 0.631. Their pointwise intervals are $[-0.939, -0.005]$ and $[0.257, 1.002]$, with signs matching equation (8). Appendix D reports alternative shock and national-state specifications. Section 5.5 treats the distinction between point estimates and uniform sign inference.

Using the same 117 policy events, the next specification conditions on six public-information variables available before the announcement. The second stage interacts these variables with the corporate and national states. The four-quarter loadings are $(-0.365, 0.664)$, with 95 percent intervals of $[-0.782, 0.056]$ and $[0.119, 1.203]$ under HC1 scaling and four-lag dependence. At radius 0.25, the fitted employment frontier becomes $(0.1205, 0.0160, 0)$. Appendix D.10 reports publication dates, the common sample, and the two-stage equivalence calculation. This exercise conditions the response contrast on the declared public-information set. Its causal interpretation continues to rely on the event-window exclusion.

The announcement-window specification includes policy-release and press-conference shocks jointly and uses an OIS basis through two years. At four quarters, the release-window loadings are $(-0.460, 0.586)$. Their Bartlett-4, HC1 intervals are $[-0.874, -0.038]$ and $[0.005, 1.170]$, with joint $p = 0.0082$. The corresponding conference block has joint $p = 0.1466$. Extending the rate factor through the two-year

OIS changes the monetary-event loading pair from $(-0.466, 0.631)$ to $(-0.469, 0.614)$. Adding orthogonal path, long-rate, and sovereign-spread components together with the six pre-announcement variables gives $(-0.404, 0.628)$. The quarter-wild intervals are $[-0.795, -0.017]$ and $[0.147, 1.107]$, and the Bartlett-4 joint value is $p = 0.0612$. Appendix D.11 reports each window, maturity basis, component, horizon, and inference calibration.

5.3. The identified response contrast

The assigned treatment is the intraday monetary innovation. Its causal interpretation rests on the event-window exclusion, the stated rate-equity restrictions, and orthogonality to country-specific outcome innovations conditional on the benchmark. The coefficients describe how the response to this treatment varies across lagged FATS states. This is a test of policy-relevant state representation with corporate control used as a measured candidate coordinate. Persistent differences in national transmission can also contribute to the response surface. Estimating the causal effect of assigning corporate control would require a separate estimand and design.

Under the legacy shock benchmark, country-specific monetary and information slopes use annual within-country index changes. The four-quarter pair is $(-0.965, 1.028)$ with intervals $[-2.594, 0.657]$ and $[-1.498, 3.646]$; the all-horizon family test gives $p = 0.721$. Adding country-specific linear trends gives $(-0.725, 0.476)$ with intervals $[-2.089, 0.632]$ and $[-1.322, 2.277]$ and family $p = 0.822$. The point directions persist across these specifications. The within-country contrasts have wide intervals.

An industry bridge matches legacy FATS states for 24 manufacturing divisions to quarterly employment. Country-by-quarter, division-by-quarter, and country-by-division effects leave comparisons among industries exposed to the same national policy environment. The foreign-control loadings are -0.090 , -0.115 , and -0.154 at horizons zero, one, and two, followed by -0.040 at four quarters. The extra-EU path is $(0.018, 0.056, 0.013, -0.034)$. Simultaneous intervals over the eight loadings include zero, and the four-quarter joint-zero value is $p = 0.873$.

An additional specification weights each cell by preceding-year manufacturing employment. At impact, the loading pair is $(-0.133, 0.104)$; both eight-coefficient simultaneous intervals exclude zero, and the joint-zero and joint one-sided sign values are 0.0001 and 0.0002 . The weighted joint values at horizons one, two, and four are 0.0137 , 0.1766 , and 0.8769 . This places the within-country evidence in the initial response. Appendix B.2 reports both designs, the country-history reassignment exercise, the national within-country specifications, and the affiliate-data requirements. Appendix C distinguishes control assignment from accompanying financing and accounting changes.

5.4. The cost of compressing the response state

Let $\ell_h = 100 \log(N_{t+h}/N_{t-1})$ for $h \in \{0, 1, 2, 4\}$ and measure error by employment RMS loss, $\|\ell\|_{\text{RMS}} = (\sum_h \ell_h^2/4)^{1/2}$. One log percentage point equals 0.01 in log employment. Stack the four unstandardized corporate-state loading rows in D_N and set $L_N = D_N/2$. The hidden state is $z = (\Delta B^F, \Delta B^E)'$ in fixed index units, while τ is an event-standard-deviation innovation at the central admissible rotation. The fitted increment is $\widehat{G}_N(z, \tau) = \tau[\widehat{R}_N(0) + L_N z]$. These choices fix the metric and scale of the approximation exercise.

The singular values of L_N are 0.4502 and 0.0661. The corresponding coordinates are $b_1 = -0.5470\Delta B^F + 0.8371\Delta B^E$ and $b_2 = 0.8371\Delta B^F + 0.5470\Delta B^E$. The first preserves 97.89 percent of the fitted squared employment-response norm. With d retained coordinates, the fitted error is $|\tau|\rho\sigma_{d+1}$; the error is zero when both coordinates are retained. At $(\rho, \tau) = (0.25, 1)$, the first coordinate lowers employment RMS error from 0.1125 to 0.0165 log percentage points. This is a reduction of 0.0960, or 85.3 percent.

The reference radius is close to the 90th percentile, 0.2436, of 178 observed same-vintage annual index changes. Because these movements also change national conditions, their scale serves only to calibrate a conditional exercise in index space. Radii 0.5 and 1 provide stress scenarios. Under the covariance ellipse $z = \Sigma_B^{1/2}u$, $\|u\| \leq 0.25$, the frontier is $(0.0603, 0.0170, 0)$. The reported frontier therefore depends on the state metric as well as the outcome weights.

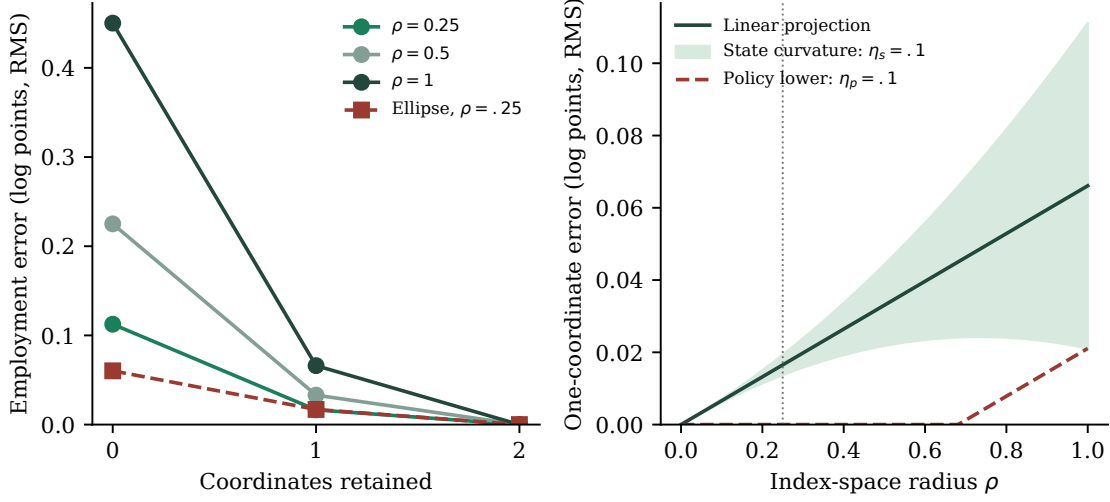
Across 315 complete EA-MPD monetary-event windows, a one-event-standard-deviation central monetary innovation moves the average factor formed from the 1-month, 3-month, 6-month, and 1-year OIS rates by 2.789 basis points. The maturity-specific impacts range from 2.568 to 3.029 basis points. At a benchmark of one million manufacturing workers, the local log-to-level derivative converts the zero-coordinate point error to 1,125 employment equivalents, with interval $[312, 1,939]$. The one-coordinate error corresponds to 165 workers, with interval $[0, 979]$, and the point reduction corresponds to 960, with aligned-score interval $[286, 1,598]$. These conversions give external units to the fitted four-horizon RMS error. A realized employment calculation would also require a state displacement, a signed response path over horizons, and the employment base at each horizon.

For a finite response map with derivative L_N at the reference state, the same radius and policy size give

$$(24) \quad \max\{0, \widehat{e}_1^{\text{lin}} - 0.03125K_s - 0.5K_p\} \leq e_1 \leq \widehat{e}_1^{\text{lin}} + 0.03125K_s + 0.5K_p,$$

where $\widehat{e}_1^{\text{lin}} = 0.25\widehat{\sigma}_{N2} \simeq 0.016523$. The units of K_s are employment RMS log points per squared index unit per policy unit; those of K_p are employment RMS log points per squared policy unit. The scenario $K_s = 0.09003$, $K_p = 0$ gives $[0.0137, 0.0193]$. Appendix D.6

FIGURE 3. Employment-Unit Repair Frontiers



The central admissible shock has unit event variance. Left: fitted frontiers for the reference radius 0.25, stress radii 0.5 and 1, and the covariance ellipse at radius 0.25. Right: one-coordinate linear error, a maintained state-curvature interval with $K_s = 0.2\hat{\sigma}_{N1}$ and $K_p = 0$, and a policy-curvature lower bound with $K_s = 0$ and $K_p = 0.2\hat{\sigma}_{N1}$. The vertical line marks the reference radius. These curvature scenarios are separate from sampling intervals.

adds metric distortion and discrepancy between the measured projection and an economic fiber. These maintained quantities govern the finite-experiment interpretation.

The aligned score draws give a 95 percent operator-norm radius of 0.3254. Weyl's bound then yields simultaneous intervals of $[0.0312, 0.1939]$ for zero-coordinate loss and $[0, 0.0979]$ for one-coordinate loss. The first interval places a positive lower bound on response loss within the measured two-index projection. Evaluating the singular-value gap directly on the same draws gives a basic interval of $[0.0286, 0.1598]$ for the point reduction of 0.0960. This functional interval is reported alongside the operator-norm envelope. Combining sampling error with the continuous shock set gives $[0, 0.1406]$ for one-coordinate loss. Curvature budgets, sampling bands, shock-set variation, and transfer from the measured projection to an economic fiber address separate parts of the exercise.

Corollary A1 states the transfer condition. With sampling radius q , the full-fiber zero-coordinate lower bound is positive when

$$(25) \quad m\hat{\sigma}_1 - Mq - \delta > \frac{\epsilon}{|\tau|\rho}.$$

For the identity measured projection, $m = M = 1$, $\delta = 0$, and $\epsilon = 0$. The remaining margin is $\hat{\sigma}_1 - q = 0.1248$ response units, which equals the 0.0312 lower endpoint after multiplication by the radius 0.25. An economy-wide statement must specify the lift constants, projection

discrepancy, and curvature remainder in equation (25). The fitted projection provides the empirical component of this inequality.

Adding credit and investment at the same horizons with fixed outcome-standard-deviation weights changes the singular values to 0.5011 and 0.2053. The second direction now accounts for 14.37 percent of squared response norm, compared with 2.11 percent for employment alone. This remains a two-column projection with its corresponding rank ceiling. The covariance-weighted KP procedure gives $p = 0.0132$. The primary Chen–Fang procedure uses an unweighted residual distance, includes the rank-zero null stratum, and retains rank one with an adjusted p -bound of 0.3848. Both calculations fit a rank-one space, so the weighting helps explain their difference. Appendix D.8 gives the full comparison. For the present exercise, the object of interest remains target-specific approximation loss together with its uncertainty.

5.5. Shock-set uncertainty and the sign target

The rotation varies over the continuous admissible shock set while the event covariance, FATS transformation, sample, and national benchmark remain fixed. The four-quarter employment point estimates range from -0.607 to -0.112 for foreign control and from 0.457 to 0.633 for extra-EU control. A one-sided intersection–union test asks whether the first loading is negative and the second positive throughout the set. It gives $p = 0.3679$. Appendix D.7 reports the component tests and the full angle sweep.

Across the retained rotations, the second singular value of the twelve-row system ranges from 0.099 to 0.334. The union band over sampling and rotation is $[0, 0.898]$. The employment-only interval in Section 5.4 uses a different target metric and is reported separately. Appendix D.5 records coefficient bands, shock scales, rank distances, and direct coefficient checks.

5.6. Financing and predictive checks

Related evidence describes several mechanisms through which joint control can affect transmission. Biermann and Huber (2024) trace a German parent-credit shock through internal funding to the sales and employment of foreign affiliates. Cetorelli and Goldberg (2012) document internal funding adjustments in global banks, and Zhang et al. (2025) connect subsidiaries’ investment responses during local monetary contractions to their access to internal capital. In my legacy-shock auxiliary analysis, credit quantities and lending rates give a centered wild-bootstrap block statistic of 65.32 and $p = 0.0005$ at the 1,999-draw resolution. The five-block Holm value is 0.0025. Appendix D.4 reports other channels at their own frequencies and under their corresponding inference rules.

An out-of-sample exercise using the alternative shock provides a separate predictive check. Training through 2018 yields a 4.71 percent reduction in employment MSE from

2019 onward. The raw gain interval spans zero, and the Clark–West value is $p = 0.0641$. Adjusted tests across chronology, FATS regime, country folds, and policy-year folds remain above 5 percent. Appendix D reports these four environments. The response frontier itself is an in-sample representation calculation, so its transport is assessed separately.

6. A Numerical Illustration of State Repair

A two-action economy illustrates what can happen when a reported state pools different decision perimeters. Its equilibrium Jacobian contains common-control curvature, a shared-input margin, strategic interaction, and an extra-EU financing buffer. Published FATS shares determine three calibration moments. Policy-response coefficients are reserved for the comparison with the model. Appendix E gives the equations, parameter mapping, and solution checks.

A matching rule selects 140 pairs using proximity in the national state and distance in the control state. Their model responses are then compared. The resulting divergence illustrates the half-separation bound for a pooled prediction. Appendix F reports model-internal errors and the model–data comparison under the legacy shock. Appendix G.4 derives state and policy curvature bounds for this economy, including control-dependent incidence, and checks the joint remainder. These constants remain in model units. The empirical frontier uses separately specified sensitivity budgets in employment units.

7. Conclusion

A national state can preserve the current allocation while pooling economies with different policy-response systems. Corporate control is one source of this variation because it joins decisions across locations. The rank of the hidden response gives the number of local coordinates required to separate those systems. Its singular values measure the loss from retaining fewer coordinates, and curvature terms extend that calculation to finite experiments.

At the central admissible shock, discarding both measured corporate coordinates leaves a positive lower bound on employment loss under the simultaneous sampling envelope. Retaining the first coordinate reduces the point loss by 85.3 percent. The value of a second coordinate depends on uncertainty in the remaining error and changes when the outcome family changes. Target weights, index-space metrics, curvature, and projection discrepancy determine how the local frontier maps into a finite economic experiment. Specifications that separate the policy-release and conference windows place the fitted response surface within the ECB announcement. The evidence concerns the representation of monetary responses across measured corporate states. The causal effect of assigning corporate control remains a separate object.

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